ADITYA MISHRA

EDUCATION

- **BS-MS in Electrical Engineering and Computer Science (EECS)** 2021 - 2026
  Indian Institute of Science Education and Research Bhopal
  CPI: 9.15/10 | **Department Rank 2** | Minor: Data Science and Engineering
- **All India Senior School Certificate Examination (AISSCE)** 2020 – 2021
  Sagar Public School, Bhopal | Percentage: 96.60

PROFESSIONAL EXPERIENCE

- **Researcher | TCS Research, Pune** July 2026 – Present
  Improving and **evaluating** the reliability of large language models on complex scientific research tasks through grounded and verifiable methods, new benchmarks, and systematic failure analysis.
  Mentor: Dr. Manasi Patwardhan, Principal Scientist, TCS Research
  Collaborator: Prof. Arman Cohan, Assistant Professor, Yale University
- **Research Intern | Fourth Signal, Mumbai** May – July 2026
  Developed a Markov chain-based probabilistic model for Expected Credit Loss (ECL) estimation on bonds and loans.

FELLOWSHIPS AND HONORS

1. **Heidelberg Laureate Forum Young Researcher 2026** Award Letter
2. **DAAD – Working Internships in Science and Engineering (WISE) 2024** Award Letter
3. **Mitacs – Globalink Research Internship 2024** Award Letter

RESEARCH EXPERIENCE

- **Master's Thesis | Low-light Signboard Recognition for Autonomous Driving** Thesis
  Curated an Indian night-time traffic dataset and **developed** a multimodal CLIP–GCNN classifier with adaptive image enhancement for traffic sign recognition. **Deployed** a lightweight TSR pipeline on NVIDIA Jetson Orin Nano with track-aware temporal stabilization and hysteresis-based label locking.
  Supervisors: Prof. Akshay Agarwal and Prof. Haroon Lone, IISER Bhopal
- **Bachelor's Thesis | Privacy-Preserving Machine Learning Models** Thesis
  Developed a mobile application incorporating a novel, personalized machine learning framework that utilized federated learning to promote students' physical well-being by analyzing their lifestyle habits.
  Supervisor: Prof. Haroon Lone, IISER Bhopal
- **Representation Learning in Electroencephalography (EEG) Data** Paper
  Proposed a graph convolutional-based variational autoencoder that leverages contrastive learning for subject representation learning from EEG data.
  Supervisors: Prof. Javad Hashemi and Prof. Ali Etemad, Queen's University Canada
- **Predicting Energy Consumption within Buildings** Paper
  Developed a state-of-the-art LSTM based framework for energy forecasting in residential and commercial buildings with limited training data leveraging physical and weather parameters.
  Supervisor: Prof. Haroon Lone, IISER Bhopal

  Additional projects (federated detection, segmentation, multi-agent RL, optimization, etc.) with code and reports Projects

PUBLICATIONS

5. Simple Features and Honest Calibration for Ambivalence and Hesitancy Recognition in Video
   Authors: Vikas Kumar, **A Mishra**, HR Lone **Status:** Accepted **Citations:** 04 **Venue:** iKDD CODS 2027
4. Beyond Benchmarks: Continuous Edge Inference for Traffic Sign Recognition on Embedded Vehicular Hardware
   Authors: A Mishra, A Tailor, HR Lone **Status:** Submitted **Venue:** ACM SenSys 2027

3. Benchmarking Nighttime Traffic Sign Recognition with Illumination-Adaptive Detection and Semantic Attribute Reasoning
Authors: A Mishra, HR Lone, A Agarwal **Status:** Submitted **Venue:** IEEE Transactions ITS

2. Subject Representation Learning from EEG using Graph Convolutional Variational Autoencoder
Authors: A Mishra, AM Samin, A Etemad, J Hashemi **Status:** Published **Citations:** 06 **Venue:** IEEE - ICASSP 2025

1. DECODE: Data-driven Energy Consumption Prediction leveraging Historical Data and Environmental Factors in Buildings
Authors: A Mishra, HR Lone, A Mishra **Status:** Published **Citations:** 72 **Journal:** Energy and Buildings

CONFERENCES AND AWARDS

5. **Heidelberg Laureate Forum Accommodation and Travel Grant 2026** Grant Letter
Selected to participate in the 13th Heidelberg Laureate Forum, Germany, connecting young researchers with Fields Medalists, Abel Prize, and Turing Award recipients. **Provided** travel grant and accommodation support by HLF Foundation.

4. **Winner of the Ambivalence/Hesitancy (AH) Video Recognition Challenge @ ECCV 2026** Certificate
Ranked first at the 11th ABAW Workshop with a reliability-gated multimodal fusion model and a calibrated ensemble.

3. **EECS Research Symposium, IISER Bhopal 2026** Poster
Showcased my MS research as a poster presentation. **Received** the Best Oral Presentation Award.

2. **Graduate Forum, IndoML 2025**
Presented my BS Thesis research. **Provided** travel, accommodation, and registration support by BITS Pilani, Hyderabad.

1. **IEEE International Conference on Acoustics, Speech, and Signal Processing 2025** *Certificate*
Chaired the session on Graph Neural Networks and delivered a **lecture** on Representation Learning on EEG data.

SKILLS

- **Languages:** Python, C, JavaScript, Mathematica
- **Data Analysis:** Pandas, Matplotlib, Seaborn
- **ML Libraries:** sklearn, Keras, TensorFlow, PyTorch
- **Tools:** Arduino, LaTeX, Stata, Bash, HPC, GitHub

EXTRACURRICULAR ACTIVITIES

1. **Athlete, Football Team** **Sports Council, IISER Bhopal**
2. **Member, Desk for Universal Design (DUDE) Lab** **UG-CREST, IISER Bhopal**
3. **Head of Logistics, Computing and Networking Council** **IISER Bhopal**

RELEVANT COURSEWORK

1. Natural Language Processing
2. Machine Learning
3. Deep Learning
4. Computer Vision
5. Applied Optimization
6. Artificial Intelligence
7. Reinforcement Learning
8. Algorithms
9. Federated Learning

ACHIEVEMENTS

5. Qualified for the Rhodes Scholarship (second-round)
4. Runner-up in Subroto Football Cup and winner in Inter IISERs Sports Meet 2022 (IISM), Football
3. Received a CSIR scholarship for outstanding performance in CBSE Class-XII examination
2. Earned scholarships for scoring a perfect 100 in CBSE Class-X Mathematics and XII Physical Education
1. Awarded a medal of distinction for securing State rank 12th in National Science Olympiad

RESEARCH INTERESTS

- **Robust perception and world models:** models that understand the 3D world and stay reliable under distribution shift.
- **Grounded and interpretable multimodal models:** vision-language models whose outputs are supported by evidence.
- **Probabilistic and reliable learning:** Bayesian and generative methods that quantify uncertainty.